

The effects of a low-fat, plant-based dietary intervention on body weight, metabolism, and insulin sensitivity.



PURPOSE: This study investigated the effect of a low-fat, plant-based diet on body weight, metabolism, and insulin sensitivity, while controlling for exercise in free-living individuals. **SUBJECTS AND METHODS:** In an outpatient setting, 64 overweight, postmenopausal women were randomly assigned to a low-fat, vegan diet or a control diet based on National Cholesterol Education Program guidelines, without energy intake limits, and were asked to maintain exercise unchanged. Dietary intake, body weight and composition, resting metabolic rate, thermic effect of food, and insulin sensitivity were measured at baseline and 14 weeks. **RESULTS:** Mean $\pm$ standard deviation intervention-group body weight decreased 5.8 ± 3.2 kg, compared with 3.8 ± 2.8 kg in the control group ($P = .012$). In a regression model of predictors of weight change, including diet group and changes in energy intake, thermic effect of food, resting metabolic rate, and reported energy expenditure, significant effects were found for diet group ($P < .05$), thermic effect of food ($P < .05$), and resting metabolic rate ($P < .001$). An index of insulin sensitivity increased from 4.6 ± 2.9 to 5.7 ± 3.9 ($P = .017$) in the intervention group, but the difference between groups was not significant ($P = .17$). **CONCLUSION:** Adoption of a low-fat, vegan diet was associated with significant weight loss in overweight postmenopausal women, despite the absence of prescribed limits on portion size or energy intake.